

SGG · AI

Smart Polyponics

One System · Zero Waste · Endless Yield

Aquaponics was only the beginning. Many production loops — fish, plants, feed and water — woven into one self-sustaining whole.



Chiang Rai, Thailand

sustainablegreengold.com

A Shared Foundation

What began as a single experiment — a small aquaponics system run as a research lab — has grown into a vision for closed-loop, self-reliant agriculture. We call it polyponics: many production loops woven into one self-sustaining whole. Our work independently arrived at the same principles championed by His Majesty King Bhumibol Adulyadej's Philosophy of Sufficiency Economy: moderation, self-reliance, resilience, and the integration of water, land and crops into a single self-sustaining system. We carry these principles forward with modern engineering.

OUR SYSTEM

Smart Polyponics

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Two Pools, One Food Chain



Premium Basin

The premium fish we're building toward

- Rainbow trout — the prized table fish we plan to raise
- Beluga sturgeon — for premium caviar, further ahead
- Clean, oxygen-rich, nutrient-managed water by design



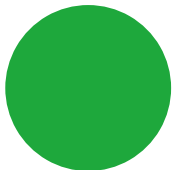
Feed Pool

A living, on-site food source

- Carp and catfish bred as natural feed
- Sustains the predatory trout & sturgeon
- Closes the chain from the very first link

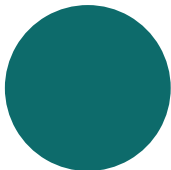
Nutrient-Rich Grow Beds

Water naturally charged with nitrates from the fish is channeled straight into the grow beds — free fertiliser that would otherwise be waste.



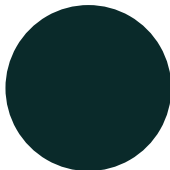
Many Crops

A wide range thrives here,
including exotic species



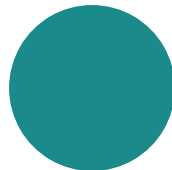
Selective Breeding

Crossbred and acclimatised to
local conditions



Fruit Trees

Mature, deep-rooted producers



Built to Adapt

Bred for strong sun and pest
resistance

SELECTIVELY BRED

Exotic Crops, Bred to Thrive

Many exotic species already grow well in our beds — our own selections, bred to be resilient and grow strongly.

FRUITS

-  Fig
-  Grapes
-  Oranges
-  Limes
-  Papaya
-  Dragon fruit
-  Date palms

HERBS & SPICES

-  Vanilla
-  Cinnamon
-  Tarragon
-  Lemongrass

PEPPERS & SPECIALTY

-  Chili peppers
-  Kratom

Our own selections — resilient, and growing strongly.

The Larvae Bioreactor



01

Feed the waste

Food scraps, spoiled meat and organic refuse are dropped into the unit.



02

Nature converts

Larvae mature on the waste, transforming refuse into protein.



03

Self-harvesting

Larvae instinctively climb a sloped ramp into a removable tray.



04

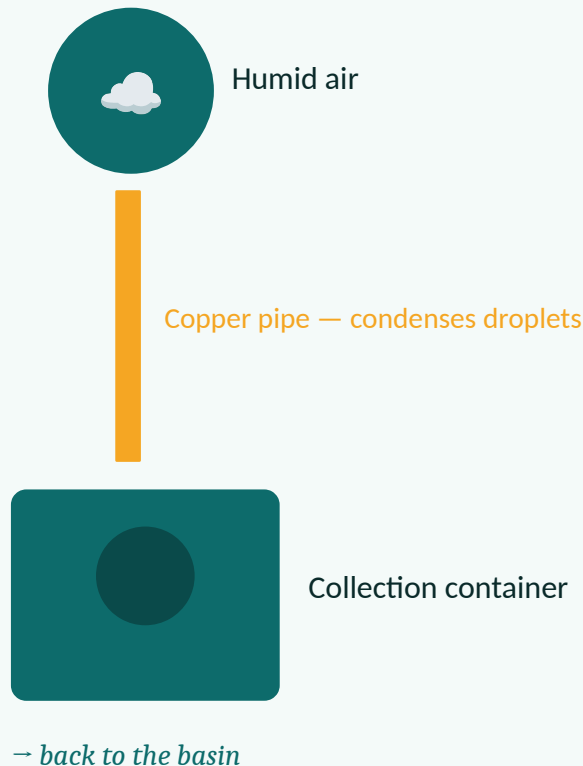
Feed the fish

Pull the tray, and clean larvae go straight to the carp.

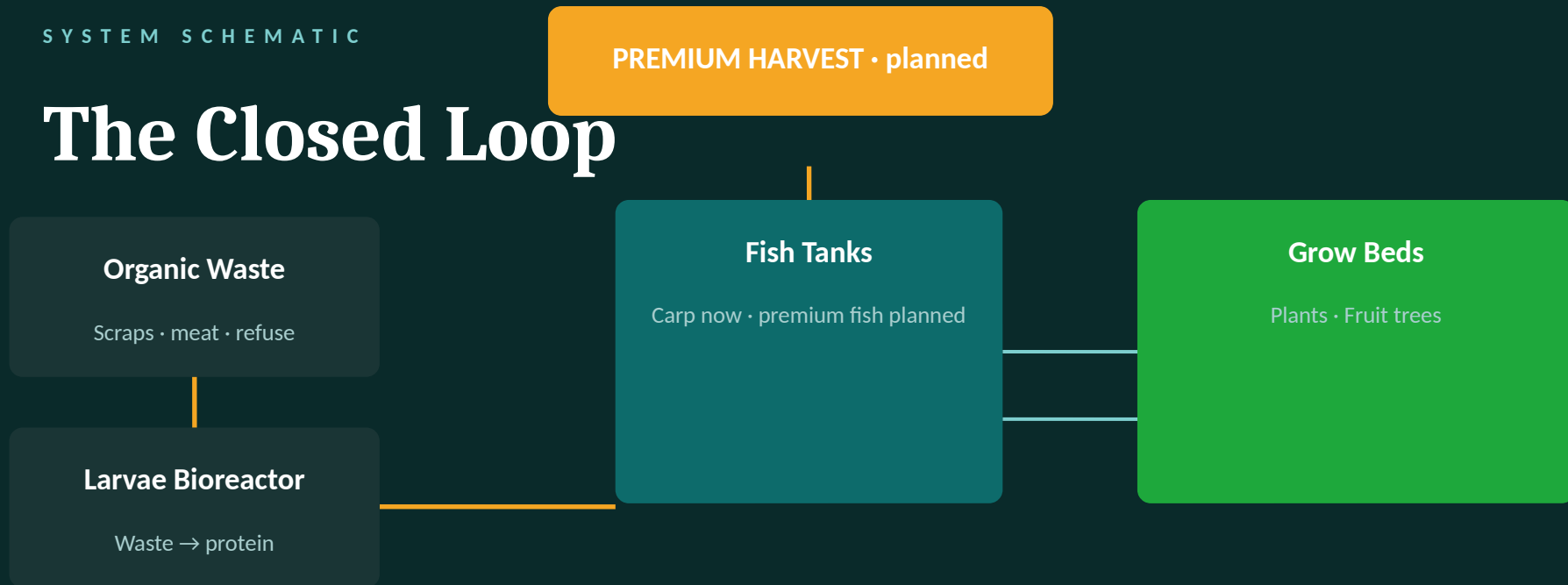
Atmospheric Water Recovery

A planned addition: the evaporation that would be lost, captured back from the air.

- A copper pipe, container and tubing will condense moisture from the humid air beside the pools.
- Harvested water collects in the container — passive, with no energy input.
- Periodically returned to the tanks to help keep the cycle whole.



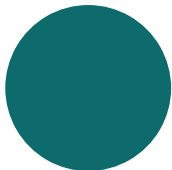
The Closed Loop



Every output becomes the next input — waste feeds larvae, larvae feed the fish, fish water feeds the plants.

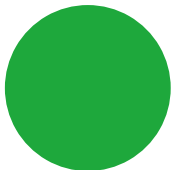
Smart Monitoring & Automation

The whole operation is built to measure itself, log everything, and increasingly run on its own.



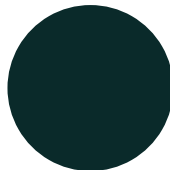
Lab-Grade Sensors

pH, dissolved oxygen, EC, ORP,
humidity and water level



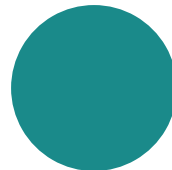
Real-Time App

Live readings, logged
continuously



AI On-Site

Runs locally on a Mac mini, at
almost no added cost

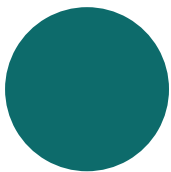


Toward Autonomy

Automated alerts and remote
control of pumps

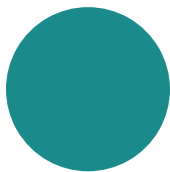
What We Want to Build Next

A complete polyponics system built from the ground up — bigger, colder where it needs to be, and largely self-running.



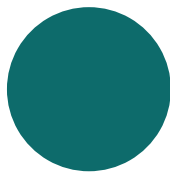
Cold-Water Tanks

Cement tanks in the ground,
kept cool



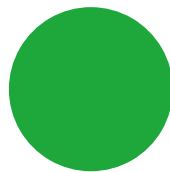
Feed Fish

Carp & tilapia as live feed



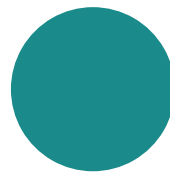
Live Protein

Larvae & frogs for premium
fish



Exotic Beds

A large area for exotic crops



Water Top-Up

Free, automatic water top-up



Full Sensing + AI

A fully instrumented build — every variable sensed, logged in our app, and watched by a built-in AI.

LOOKING AHEAD

Built for the 15-Minute City

Cities everywhere are reorganising around one idea: everything you need within a short walk or ride. Food is the piece still missing — most of it travels long distances to reach the table.

A compact, self-sufficient polyponics system changes that — growing fresh fish and produce right where people live. That is the future we are building toward.



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